

MULTIMEDIA TECHNOLOGY

Unit-2: Sound in Multimedia

Comprehensive Study Notes

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Quick Reference Summary

1. Sound and Its Attributes

Definition

Sound is a vibration that propagates as an acoustic wave through a transmission medium such as air, water, or solids. In multimedia, sound is a crucial element that enhances user experience and engagement.

Nature of Sound

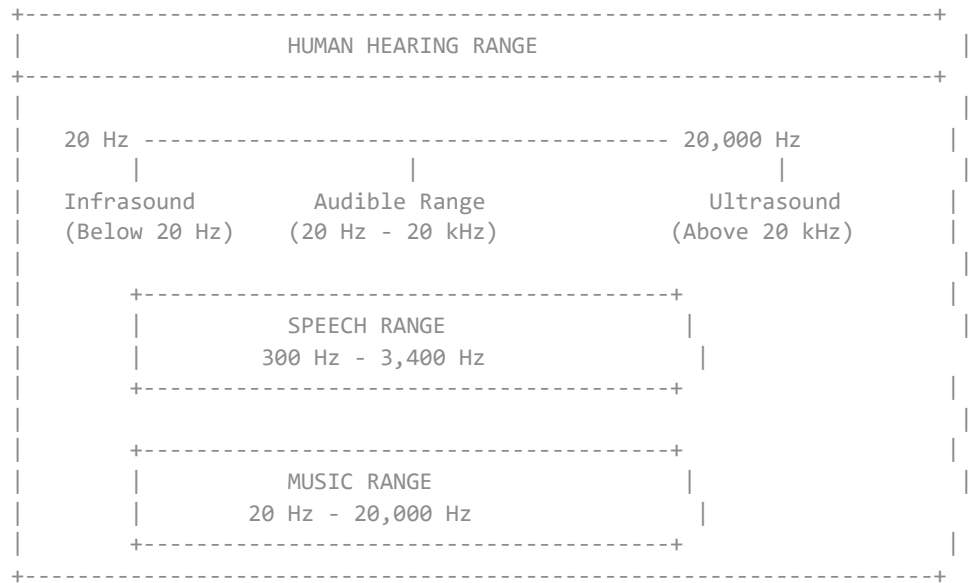
Sound is created by vibrations that create pressure waves in the air. These waves are detected by our ears and interpreted by our brain.

PHYSICAL NATURE OF SOUND

Sound Attributes Table

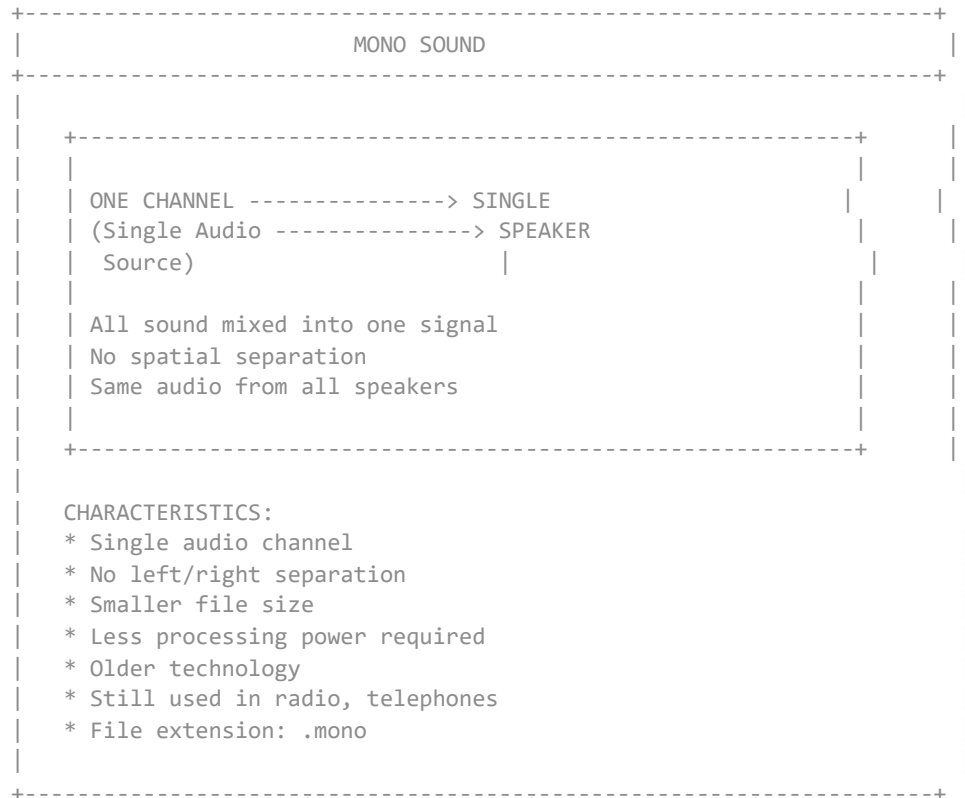
Attribute	Description	Measurement	Effect
Amplitude	Height of sound wave	Decibels (dB)	Determines volume/loudness
Frequency	Number of cycles per second	Hertz (Hz)	Determines pitch
Wavelength	Distance between wave peaks	Meters	Affects sound propagation
Timbre	Tone quality/color	Subjective	Distinguishes instruments
Duration	Length of sound	Seconds	Defines sound length
Phase	Wave position in cycle	Degrees	Affects sound interference
Velocity	Speed of sound wave	m/s (343 m/s in air)	Affects timing

Human Hearing Range

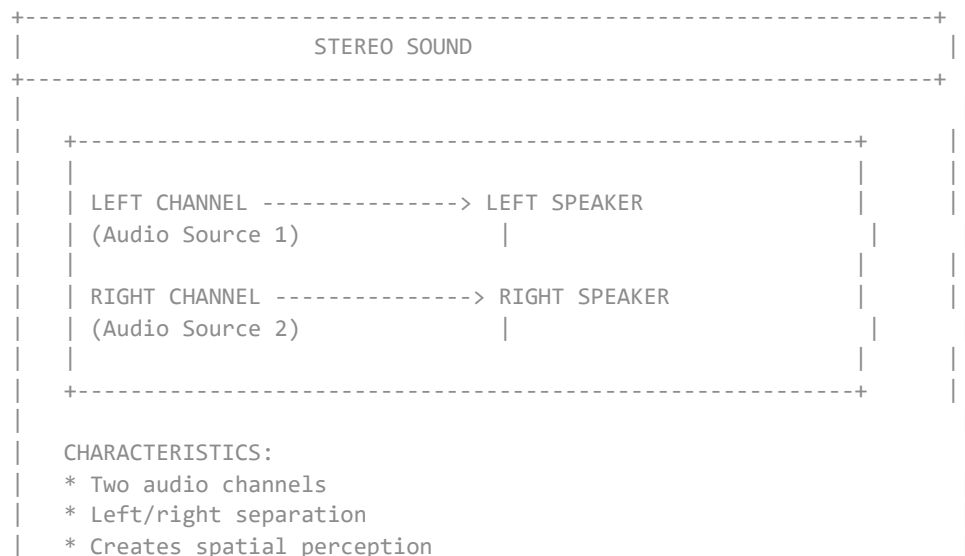


2. Mono vs Stereo Sound

Mono Sound (Monophonic)



Stereo Sound (Stereophonic)



- * Gives directionality to sound
- * Larger file size (approx 2x mono)
- * More processing power required
- * Standard for music, movies
- * Creates immersive experience

Mono vs Stereo Comparison

Feature	Mono Sound	Stereo Sound
Channels	1 channel	2 channels
Spatial Effect	No spatial separation	Left/Right separation
Directionality	No directional cues	Directional cues present
File Size	Smaller	Larger (2x mono)
Processing	Less intensive	More intensive
Sound Quality	Flat, centered	Rich, spacious
Equipment Needed	Single speaker	Two speakers
Best Used For	Voice, radio, phones	Music, movies, games
Cost	Less expensive	More expensive
Immersion	Low	High

3. Sound Channels

Concept of Sound Channels



Detailed Channel Configurations

Configuration	Channels	Speaker Layout	Application
Mono (1.0)	1	Single center	Radio, Voice
Stereo (2.0)	2	L + R	Music, TV
2.1	3	L + R + Subwoofer	PC speakers
4.1	5	Front L/R + Rear L/R + Sub	Gaming
5.1	6	Front L/R + Rear L/R + Center + Sub	Home theater
6.1	7	5.1 + Rear Center	Movies
7.1	8	5.1 + Side L/R	Advanced theater
Dolby Atmos	Variable	Multiple overhead + floor	Cinema, VR

4. Sound and Its Effect in Multimedia

Role of Sound in Multimedia

EFFECTS OF SOUND IN MULTIMEDIA	
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Types of Sound in Multimedia

Type	Purpose	Examples
Background Music	Sets mood, atmosphere	Instrumental tracks
Narration/Voice-over	Explains content	Educational videos
Sound Effects (SFX)	Enhances realism	Footsteps, explosions
Ambient Sound	Creates environment	Rain, traffic, wind
Dialogues	Communication	Character speech
User Interface Sounds	Feedback	Clicks, notifications
Audio Cues	Signals actions	Alerts, warnings

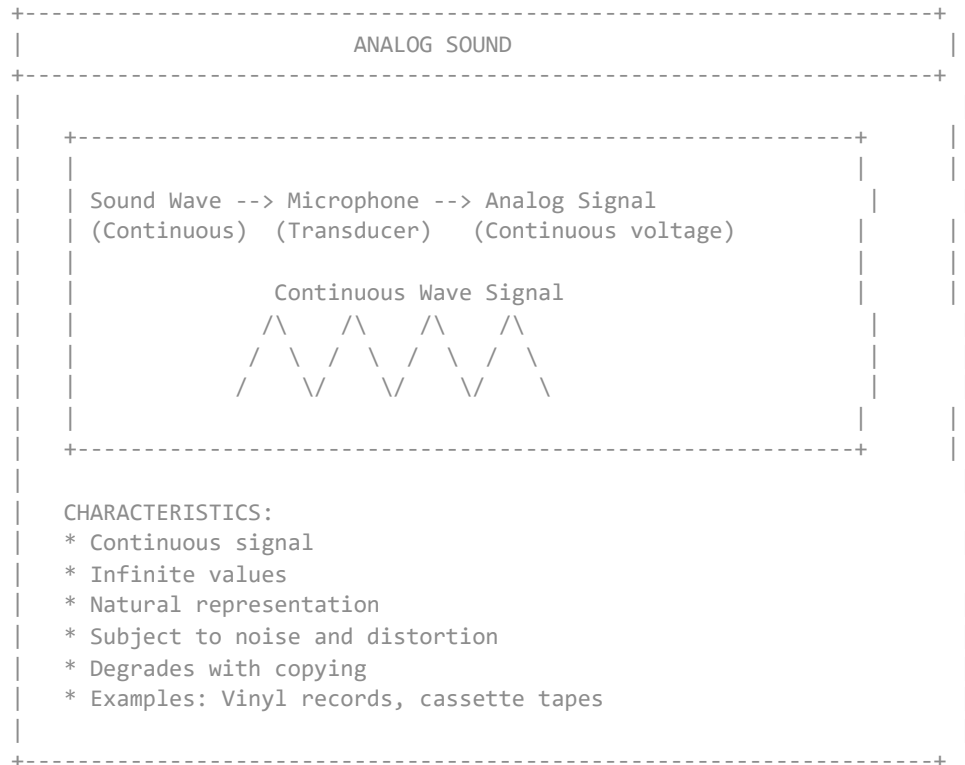
Psychological Effects of Sound

PSYCHOLOGICAL EFFECTS OF SOUND	
SOUND TYPE	PSYCHOLOGICAL EFFECT
Slow tempo music	Relaxation, calmness
Fast tempo music	Excitement, energy
Minor key music	Sadness, melancholy
Major key music	Happiness, joy
Dissonant sounds	Tension, discomfort
Consonant sounds	Harmony, comfort
White noise	Focus, concentration
Sudden loud sounds	Alert, startled
Nature sounds	Peaceful, natural
Voice pitch variation	Emotion, meaning

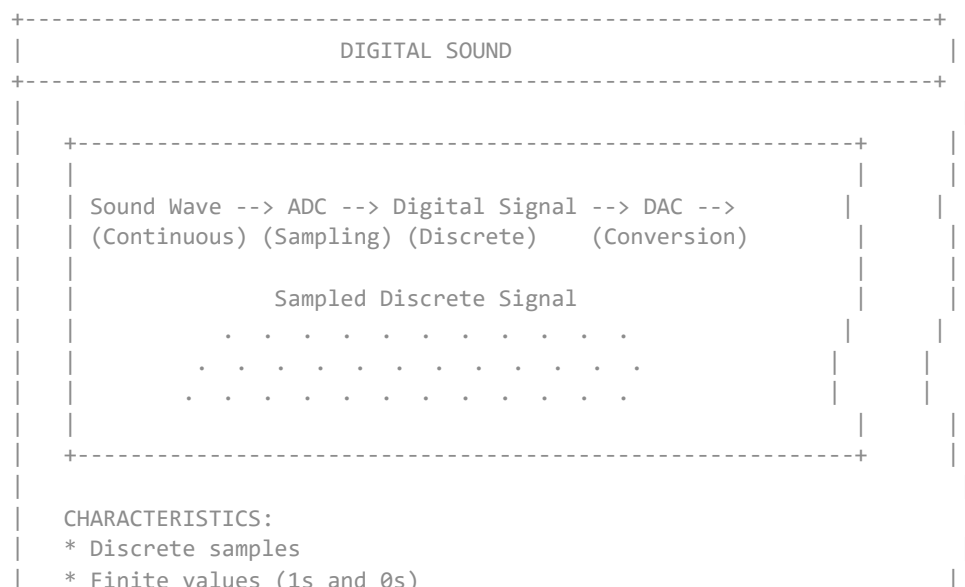


5. Analog vs Digital Sound

Analog Sound



Digital Sound

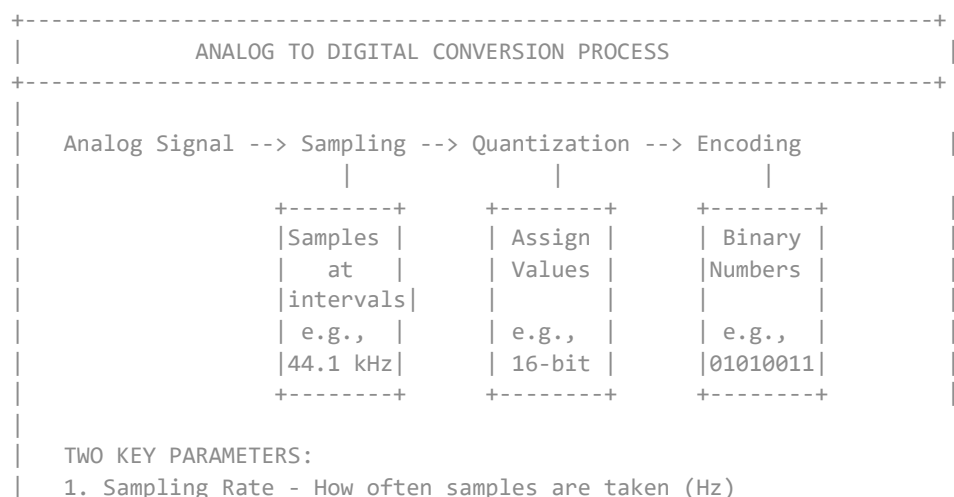


- * Converted from analog
- * No degradation on copying
- * Resistant to noise
- * Examples: CD, MP3, WAV files

Analog vs Digital Comparison

Feature	Analog Sound	Digital Sound
Signal Type	Continuous wave	Discrete samples
Values	Infinite	Finite (binary)
Quality	Original, natural	Depends on sampling
Noise	Susceptible to noise	Noise-resistant
Copying	Degrades with copying	Perfect copies
Storage	Physical media	Electronic files
Editing	Difficult	Easy
Format	Cassette, Vinyl	WAV, MP3, CD
Cost	Generally cheaper	Higher initial cost
Durability	Fragile	Durable

ADC (Analog to Digital Conversion) Process



2. Bit Depth - Number of bits per sample (resolution)

Sampling Quality Comparison

Quality	Sampling Rate	Bit Depth	Applications
Telephone	8 kHz	8-bit	Voice calls
AM Radio	11 kHz	8-bit	Radio broadcast
FM Radio	22 kHz	16-bit	Radio broadcast
CD Quality	44.1 kHz	16-bit	Music CDs
DVD Quality	48 kHz	24-bit	Movies
Studio Quality	96 kHz	24-bit	Professional recording
High Resolution	192 kHz	32-bit	Advanced mastering

6. Overview of Various Sound File Formats

Sound File Formats Classification

+	-----+
	SOUND FILE FORMATS CLASSIFICATION
+	-----+

6.1 WAV (Waveform Audio File Format)

+	-----+
	WAV FILE FORMAT
+	-----+
	FULL NAME: Waveform Audio File Format
	DEVELOPED BY: Microsoft & IBM
	FILE EXTENSION: .wav
	TYPE: Uncompressed
	WAV FILE STRUCTURE:
	+
	HEADER (44 bytes)
	- File type (RIFF)
	- File size
	- Audio format
	- Sample rate (e.g., 44.1 kHz)
	- Bit depth (e.g., 16-bit)
	- Number of channels (mono/stereo)
	- Data size
	DATA
	- Raw audio samples
	- PCM (Pulse Code Modulation) data
	CHARACTERISTICS:
	* Uncompressed audio
	* High quality
	* Very large file size
	* Ideal for professional use
	* Standard on Windows platform
	* Widely supported
	WAV FILE SIZE CALCULATION:
	File Size = Sample Rate x Bit Depth x Channels x Duration
	Example: CD Quality (44.1 kHz x 16-bit x 2 x 60 seconds)
	= 44,100 x 2 x 2 x 60 = 10.5 MB per minute
+	-----+

6.2 MP3 (MPEG Audio Layer 3)

MP3 FILE FORMAT

FULL NAME: MPEG-1 Audio Layer 3

DEVELOPED BY: Fraunhofer Society

FILE EXTENSION: .mp3

TYPE: Compressed (Lossy)

MP3 COMPRESSION PROCESS:

Original Audio	--> Filter	--> Quantize	--> Bit
(WAV)	(PSYCHO-ACOUSTIC MODEL)	(Reduce Data)	(MP3)

Uses Psychoacoustic Model:

- Removes inaudible sounds
- Masking effect (loud sounds hide soft ones)
- Reduces file size by 90%

CHARACTERISTICS:

- * High compression ratio
- * Good quality at high bit rates
- * Variable and constant bit rates
- * Most popular format for music
- * Universal support
- * Small file size

MP3 BIT RATES AND QUALITY:

BIT RATE	QUALITY	USE CASE
320 kbps	Excellent	Audio CDs, Archiving
256 kbps	Very Good	High quality music
192 kbps	Good	Standard music
128 kbps	Acceptable	Streaming, Portable
64 kbps	Poor	Voice only

FILE SIZE COMPARISON (3 minutes):

WAV (CD Quality) = 30 MB

MP3 (128 kbps) = 2.8 MB

MP3 (192 kbps) = 4.2 MB

MP3 (320 kbps) = 7.0 MB

6.3 Other Sound File Formats

SOUND FILE FORMATS COMPARISON TABLE					
FORMAT	TYPE	QUALITY	SIZE	USE CASE	
WAV	Uncomp.	Excellent	Very Large	Production	
AIFF	Uncomp.	Excellent	Very Large	Mac Audio	
MP3	Lossy	Good	Small	Music, Web	
AAC	Lossy	Better	Small	iTunes, Web	
OGG	Lossy	Good	Small	Open Source	
WMA	Lossy	Good	Small	Windows	
FLAC	Lossless	Excellent	Medium	Archiving	
ALAC	Lossless	Excellent	Medium	Apple Music	
PCM	Uncomp.	Excellent	Very Large	Recording	
MIDI	Control	Depends	Very Small	Synthesizer	

Detailed Format Descriptions

Format	Description	Advantages	Disadvantages
WAV	Windows standard format	High quality, no loss	Large file size
AIFF	Apple standard format	High quality, Mac native	Large file size
MP3	Most popular compressed format	Small size, universal	Lossy compression
AAC	Advanced Audio Coding	Better quality than MP3	Less universal
OGG	Open source format	Free, good quality	Limited support
WMA	Windows Media Audio	Small size, Windows support	Platform dependent
FLAC	Free Lossless Audio Codec	Lossless, smaller than WAV	Less universal
ALAC	Apple Lossless	Lossless, Apple ecosystem	Limited to Apple devices
MIDI	Musical Instrument Digital Interface	Extremely small, editable	Not actual audio

6.4 MIDI (Musical Instrument Digital Interface)

MIDI FILE FORMAT			
FULL NAME: Musical Instrument Digital Interface			
FILE EXTENSION: .mid, .midi			
TYPE: Control/Instruction format			
MIDI CHARACTERISTICS:			
* Does NOT contain audio data * Contains instructions for instruments * Data includes: - Note On/Off events - Pitch - Velocity (loudness) - Instrument selection - Duration			
CHARACTERISTICS:			
* Very small file size * Needs synthesizer to play * Editable at note level * Used in karaoke, mobile ringtones * Supports 16 channels per device * 128 standard instruments (GM)			
MIDI VS AUDIO COMPARISON:			
MIDI:	+-----+ +-----+		
File	Instructions	--> Synthesizer	--> Sound
	+-----+ +-----+		
AUDIO:	+-----+ +-----+		
File	Actual Sound	--> Speakers	--> Sound
	Data		
	+-----+ +-----+		

Visual Learning: Flowcharts

Flowchart: Sound Format Selection Guide

SOUND FORMAT SELECTION GUIDE	
RECOMMENDATION TABLE	
PURPOSE	RECOMMENDED FORMAT
Professional Recording	WAV, FLAC
Music Listening	MP3 (192-320 kbps), AAC
Podcasts	MP3 (128-192 kbps)
Web Streaming	MP3 (128 kbps), OGG
Mobile Devices	AAC, MP3
Archiving	FLAC, ALAC
Voice Recording	WAV, MP3 (64 kbps)
Video Production	WAV, AIFF
Ringtone	MIDI, MP3 (low bitrate)
Karaoke	MIDI

Flowchart: Audio Editing Workflow

AUDIO EDITING WORKFLOW		
TRIMMING	NOISE REDUCTION	NORMALIZATION
Remove Silence	Remove Hiss	Volume Levelling
Cut Sections	Background	Peak/Average
Crop Audio	Filtering	Adjustment
EQUALIZATION	EFFECTS	MULTI-TRACK
Frequency	Reverb	Mixing
Adjustment	Echo	Layering
Filtering	Chorus	Synchronization

Quick Reference Summary

Key Terms

1. **Amplitude** - Volume/loudness of sound
2. **Frequency** - Pitch of sound (Hz)
3. **Sampling Rate** - Samples per second (Hz)
4. **Bit Depth** - Bits per sample (resolution)
5. **Mono** - Single audio channel
6. **Stereo** - Two audio channels
7. **Lossy Compression** - Removes data to reduce size
8. **Lossless Compression** - Maintains quality, reduces size
9. **Psychoacoustics** - Study of sound perception
0. **ADC** - Analog to Digital Conversion

Sound Channels Quick Reference

Mono (1.0)	--> Single speaker
Stereo (2.0)	--> L + R speakers
5.1	--> L + R + Center + Surround + Sub
7.1	--> 5.1 + Side speakers
Dolby Atmos	--> Adds overhead channels

Sound File Size Formula

File Size = Sample Rate x Bit Depth x Channels x Duration

Questions

Q1: Define sound and list its attributes. Explain each attribute with its measurement unit.

Q2: Differentiate between mono and stereo sound with suitable examples.

Q3: Explain various sound channel configurations (Mono, Stereo, 5.1, 7.1, Dolby Atmos).

Q4: Describe the role and psychological effects of sound in multimedia applications.

Q5: Compare analog and digital sound in detail. Explain the ADC process.

Q6: What is sampling rate and bit depth? Explain their significance with a quality comparison table.

Q7: Explain the WAV file format including its structure and size calculation.

Q8: Describe the MP3 compression process using the psychoacoustic model.

Q9: Compare various sound file formats (WAV, MP3, AAC, FLAC, OGG, WMA, AIFF).

Q10: What is MIDI? Explain how it differs from digital audio formats.

Note: These notes cover Unit II - Sound in Multimedia as per HPU BCA Semester VI curriculum. Review all topics thoroughly for examinations.

— End of Unit II Notes —